SpectraForge: An Open, Physics-Grounded Benchmark for Band Selection in Multi-Excitation (4D) Fluorescence Hyperspectral Imaging

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Multi-excitation fluorescence hyperspectral imaging records, for every pixel, a four-dimensional excitation-emission cube, and choosing which spectral bands to keep is central to cheaper acquisition and compact classification. Progress on band selection for this modality is held back by a data problem: labeled real 4D fluorescence cubes are scarce, and even when available they carry no ground truth, so a selector cannot be scored against the bands it should have found. We present **SpectraForge**, an open, physics-grounded generator that renders 4D excitation-emission cubes from a parametric fluorophore model obeying the standard laws of fluorescence photophysics (Kasha, Vavilov, Franck-Condon, Stokes shift), with optional inner-filter attenuation, reabsorption, FRET, scatter, photon and read noise, and fixed-pattern clutter. Every cube carries perfect ground truth (concentration maps and true emission peaks), parametric dyes can be replaced by real measured spectra imported from FPbase, and an interactive workbench lets a user paint scenes, render, and validate a selection against ground truth. Crucially, scenes are built so that band variance is decoupled from class informativeness (measured correlation ≈ -0.1), which makes the benchmark an honest test rather than a trivial one, at difficulty controllable from a single call.

Alongside the generator we define an evaluation protocol whose two decisive controls, a random-selection baseline and the full spectrum, separate genuine gains from self-deception, together with a classifier panel and a nonlinear-only information metric for regimes where linear scoring cannot credit a nonlinear-shape selection.

Using SpectraForge we characterize autoencoder-plus-latent-perturbation band selection. The benchmark exposed a failure invisible to raw loss (a published model that selected at chance, macro-F1 ≈ 0.33) and traced it to four trainer defects and an emission-axis band collapse; correcting these recovers the selector to 85–97% of a labels-using oracle and matches the strongest classical blind selector (PCA loadings). A significance-tested battery then shows the learned selector *ties* PCA loadings on realistic and FRET data and *significantly exceeds* it in the nonlinear reabsorption regime (macro-F1 0.449 vs 0.431, 95% CI $[+0.001, +0.029]$), capturing nonlinear-only information that linear selection misses. The same controls delimit where unsupervised selection stops helping, and reveal that a convolutional model is structurally necessary when discrimination is carried by spatial texture, where per-pixel selection is provably blind (below random) while a spatial model reaches the oracle.

SpectraForge turns "does this selector help" into a measured, reproducible question for 4D fluorescence imaging. The generator, the evaluation harness, and the full experiment record are released, and validation on real Lichens and Collagen cubes with the same controls is the immediate next step.